

Patterns of Crime and Arrest Disparities in Tucson: A Multivariate Perspective

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Chapter 1

Introduction

In this project, we are studying how environmental and socioeconomic factors affect crime and arrest patterns in neighborhoods across Tucson, Arizona. Crime is not evenly spread throughout a city, and some areas may experience more police activity or higher arrest rates than others. We want to understand if these differences are related to things like income, housing values, or access to infrastructure such as street lighting.

Our main goal is to find out whether certain neighborhood conditions are linked to higher crime or arrest rates. We also want to see if there are patterns in the data that suggest some areas are policed more heavily than others, even when crime levels are similar. These questions are important for making sure public safety policies are fair and based on evidence, not just assumptions.

1.1 Objectives

To study this, we are using public datasets from the City of Tucson. These include records of reported crimes and arrests, information about neighborhood income levels and housing values, and data about streetlights and other infrastructure. By combining these sources, we hope to conduct a comparative analysis between these different. Our research questions are as follows:

- Do thefts and violent crimes occur more often in richer or poorer neighborhoods?
- Does the existence or presence of city streetlights influence crime rates?

1.2 Overview of Data

This project uses multiple datasets to explore how socioeconomic and environmental conditions are related to crime and arrest disparities in Tucson neighborhoods. Each dataset supports a specific part of our analysis, allowing us to examine relationships between location-based factors, community demographics, and law enforcement activity.

1. Tucson Police Reported Crimes

This dataset provides detailed records of reported crimes within the City of Tucson. Each entry includes the type of crime, the location, and the date and time of the report. This data will be used to analyze spatial and temporal patterns of criminal activity at the neighborhood level.

2. Tucson Police Arrests

The arrests dataset contains information on individual arrests made by Tucson Police, including the offense type, arrest location, and demographic details of the individuals arrested. This dataset is central to our analysis of disparities in arrest rates across neighborhoods with different socioeconomic profiles.

3. City of Tucson Streetlight Locations

This dataset contains the locations of public streetlights across Tucson. We will use it to examine whether nighttime visibility correlates with crime or arrest patterns. It will also support our environmental analysis by identifying areas with limited lighting.

4. Neighborhood Income

Neighborhood-level income data is drawn from a publicly available CSV that aggregates income estimates, likely based on census tract boundaries. This data will help us evaluate the economic conditions of each area and investigate how they relate to both crime rates and arrest disparities.

1.3 Related Works

- **Effects of Improved Street Lighting on Crime** by Welsh, B.C. and Farrington, D.P. (2008) [\[WF08\]](#)

This study examines the impact of improved street lighting on crime rates, finding that better lighting can reduce certain types of crime. This research informs our hypothesis about the relationship between streetlight presence and crime rates in Tucson.

- **Income Inequality, Race, and Place: Does the Distribution of Race and Class Within Neighborhoods Affect Crime Rates?** by Hipp, J.R. (2007) [\[Hip07\]](#)

This study investigates the relationship between income inequality, race, and neighborhood crime rates. Hipp finds that income disparities within neighborhoods significantly influence crime, supporting our analysis of socioeconomic factors in Tucson.

1.4 Methods and Techniques

This section covers how the four datasets were cleaned and merged, which models were fitted to the result, and how those models were validated.

1.4.1 Data Preprocessing and Integration

- **Data Cleaning and Standardization**

Four datasets were used: Tucson Police Reported Crimes, Tucson Police Arrests, City of Tucson Streetlight Locations, and Neighborhood Income. Pandas, NumPy, and GeoPandas loaded and processed them. Key steps included converting DateOccurred to datetime, categorizing crimes based on TimeOccur (e.g., Night), and removing rows with missing values in critical fields like Ward.

- **Feature Standardization**

Ward values were standardized as integers across datasets. Arrest records were filtered for valid coordinates and Ward information. Streetlight data was limited to active lights with numeric Wattage. Income data was refined to include only WARD, MEDHINC_CY, and AVGHINC_CY.

- **Dataset Merging**

Crime and arrest counts were aggregated by ward, and a new feature called Night_Crime_Prop was calculated to represent the proportion of nighttime crimes. These aggregates were merged with income data using WARD, and streetlight data was incorporated through spatial joins.

1.4.2 Statistical and Machine Learning Models

- **Ordinary Least Squares (OLS) Regression**

Used to examine the relationship between Crime_Count and predictors such as MEDHINC_CY and Streetlight.Count. Implemented using the statsmodels library.

- **Random Forest Classifier**

Used to predict high-crime wards based on features including MEDHINC_CY, AVGHINC_CY, and Streetlight.Count. Implemented with scikit-learn and addressed class imbalance using SMOTE (Synthetic Minority Over-sampling Technique).

1.5 Evaluation Framework

- **Cross-Validation**

K-Fold cross-validation was used to evaluate model generalizability across multiple subsets of the data, minimizing overfitting and ensuring consistent performance.

- **Performance Metrics**

Accuracy, precision, recall, F1-score, ROC AUC, mean squared error (MSE), and R^2 were computed for each model.

- **Statistical Modeling and Testing**

Statsmodels was used to test which predictors were significant and to check the regression assumptions.

- **Visualization**

Both static and geospatial visualizations were used to communicate findings effectively. Libraries such as Matplotlib and Seaborn were leveraged to create interpretable visual summaries of trends and spatial distributions.

1.6 Software and Tools

1.6.1 Python

was the primary language, used for data processing, modeling, and visualization. Key libraries included

- Pandas and NumPy for efficient data manipulation and numerical operations
- Matplotlib and Seaborn for static data visualization.
- Geopandas, Shapely, and Geopy for handling and visualizing geospatial data
- Scikit-learn and Imbalanced-learn for implementing machine learning models, performing oversampling (SMOTE), and conducting model evaluation using metrics such as accuracy, F1-score, and ROC AUC
- Statsmodels for conducting statistical modeling and significance testing

1.6.2 Colab Notebook

Google Colab was the development platform. It runs the notebook in the cloud and keeps the code, its plots, and the notes on both in one file, so a collaborator opens the analysis and its output together.

Chapter 2

Exploratory Data Analysis

This section examines the relationships between crime types, wards, and time periods using stacked bar charts to uncover patterns that may inform hypotheses about crime distribution and potential mitigation strategies.

2.1 Crime Types by Ward

Figure 2.1 illustrates the distribution of various crime types across the city's six wards, labeled 1 through 6. The crime categories include Homicide, Sexual Assault, Robbery, Aggravated Assault, Burglary, Larceny, Grand Theft Auto (GTA), and Arson. A key observation is that Larceny (represented by the largest brown segment) is the most frequent crime type in every ward, consistently accounting for the majority of incidents.

For example, Ward 3 reports the highest crime count, approaching 25,000 incidents, with Larceny contributing the largest share. In contrast, Ward 4 has the lowest crime count, around 5,000 incidents, yet Larceny remains the predominant crime there as well. Less common crime types such as Homicide (blue) and Arson (gray) appear in very small proportions across all wards, indicating their rarity.

Violent crimes such as Aggravated Assault (red) and Robbery (green) are more prevalent in Wards 3 and 5, the two highest-crime wards. That points at ward-level characteristics: income, housing density, or another socioeconomic factor. It supports Hypothesis 1, that violent and property crimes cluster by income profile.

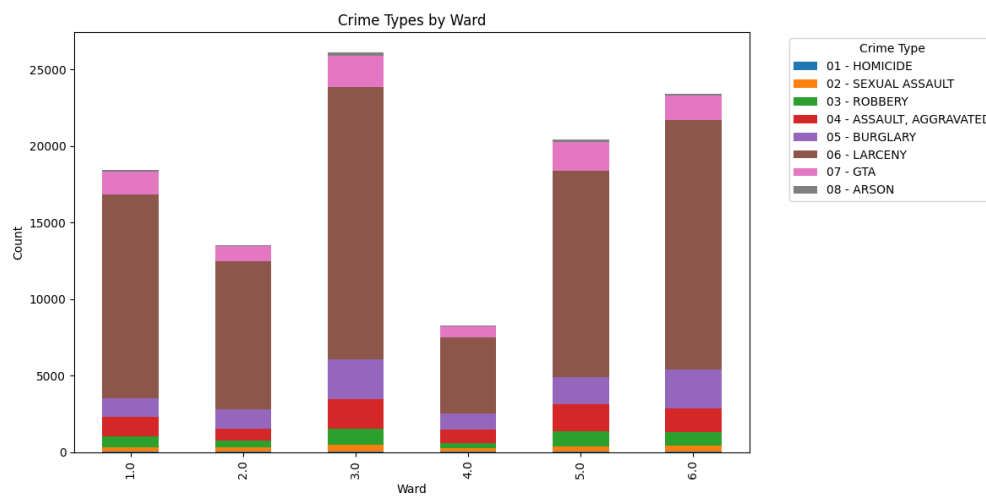


Figure 2.1: Prevalence of Crime Type per Ward

2.2 Crime Types by Time

The second stacked bar chart, “Crime Types by Time Period” (Figure 2.2), categorizes incidents into four daily time periods: Morning, Afternoon, Evening, and Night. Consistent with the ward-based analysis, Larceny again emerges as the most common crime in every time period. The Afternoon period shows the highest overall crime count, exceeding 35,000 incidents, while Nighttime reports the lowest, at approximately 15,000.

Although violent crimes such as Robbery and Assault account for a slightly larger share of incidents at Night compared to other times, their overall numbers remain relatively low. This suggests that while violent crimes are somewhat more likely to occur during nighttime hours, they do not significantly shift the overall crime landscape.

The high incidence of Larceny during the Afternoon and Evening may be linked to greater public activity during those hours, which increases opportunities for theft. In terms of mitigation, these findings suggest that increasing streetlight density could specifically help deter nighttime Robbery and Assault, though such efforts may have limited impact on overall crime, given Larceny’s dominance and its timing.

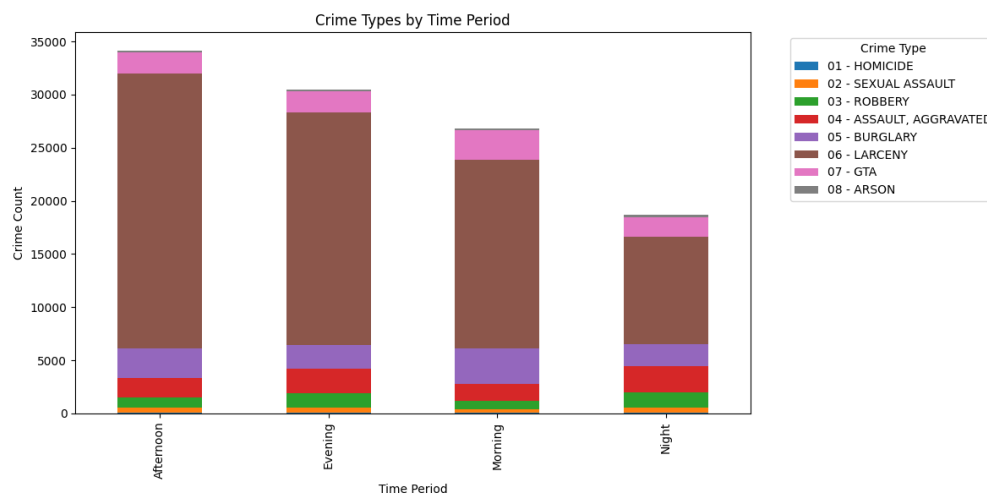


Figure 2.2: Prevalence of Crime Type by Time Period

To further refine this temporal analysis, the histogram “Crime Count by Hour of Day” in Figure 2.3, provides a granular view of crime distribution across the 24-hour cycle. Crime counts stay below 2,000 incidents from midnight to 6 AM. They climb sharply from about 7 AM and peak between 10 AM and 2 PM at over 6,000, which is where Larceny and other opportunistic crimes concentrate. The count falls after 2 PM but stays high into the evening, with a second peak from 6 PM to 8 PM that matches the Evening period’s Larceny rate. After 10 PM it drops to the Night totals. The kernel density curve gives the same bimodal shape. On these numbers, prevention effort belongs in the 10 AM to 8 PM window, and streetlighting is a night measure aimed at what is left after 10 PM.

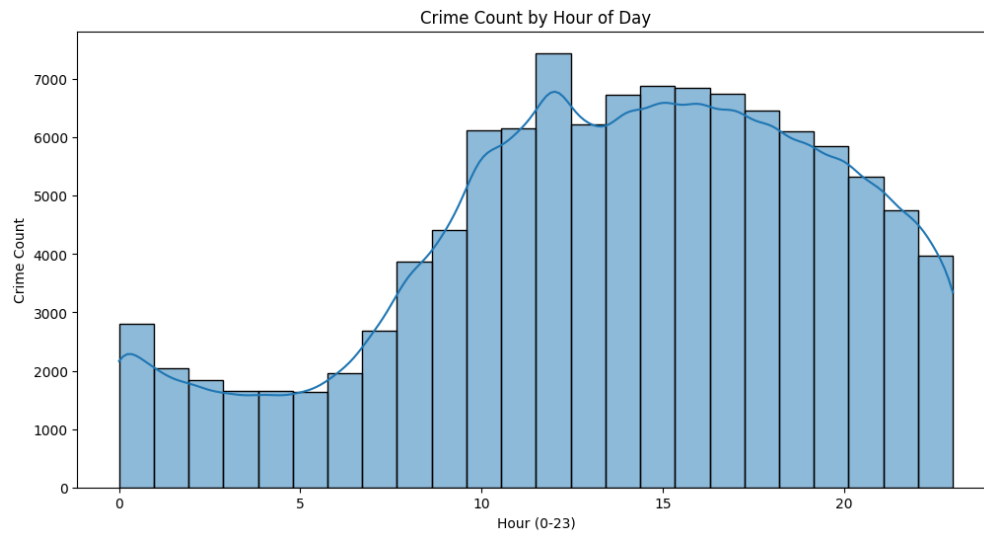


Figure 2.3: Total Crimes per Hour Throughout the 24 Hour Period

2.3 Correlation Matrix Heatmap

The heatmap provided in Figure 2.4 is a correlation matrix that illustrates the relationships between various variables: median household income (MEDHINC_CY), average household income (AVGHINC_CY), streetlight count (Streetlight_Count), crime count (Crime_Count), arrest count (Arrest_Count), and the proportion of nighttime crimes (Night_Crime_Prop). The color intensity and values (ranging from -1.0 to 1.0) indicate the strength and direction of the correlation, with positive values showing a direct relationship and negative values indicating an inverse relationship.

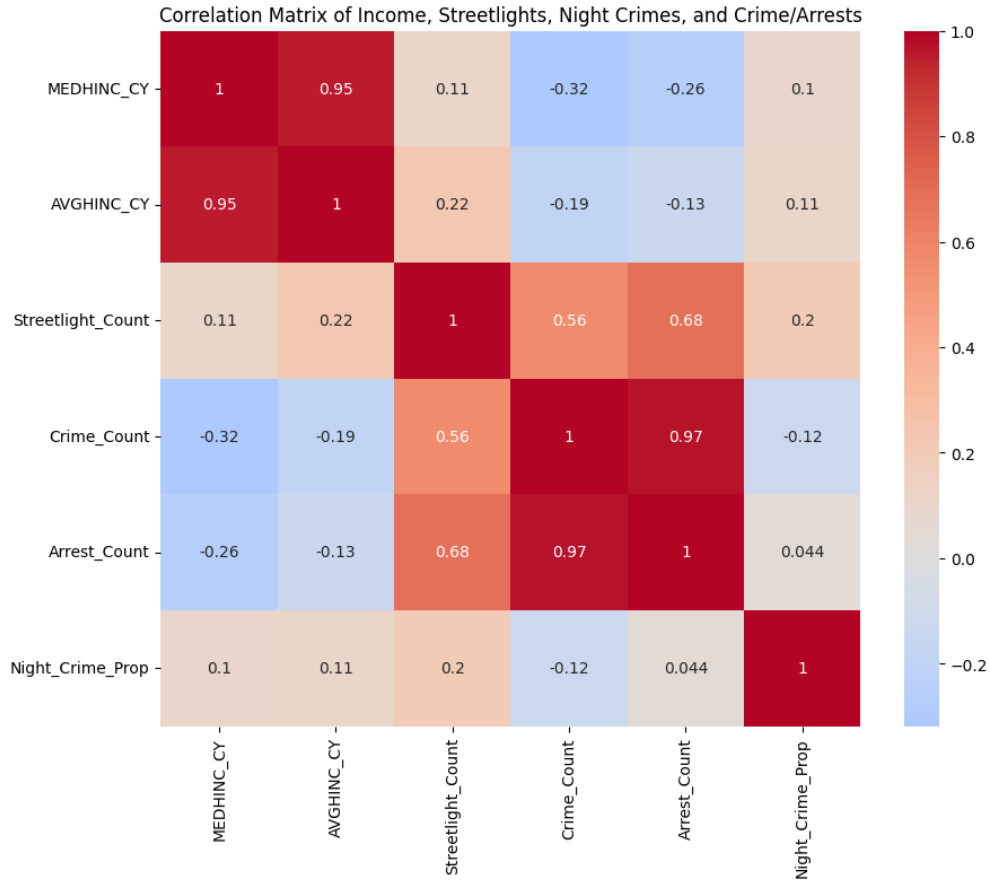


Figure 2.4: Correlation Matrix Heatmap

2.3.1 Key Insights

- **Income vs. Crime and Arrests**

MEDHINC_CY and AVGHINC_CY exhibit a strong positive correlation with each other (0.95), indicating that median and average incomes in neighborhoods are highly consistent. Both MEDHINC_CY and AVGHINC_CY have moderate negative correlations with Crime_Count (-0.32 and -0.19 , respectively) and Arrest_Count (-0.26 and -0.13 , respectively). This suggests that higher income levels are associated with lower crime and arrest rates, though the relationship is not extremely strong.

- **Streetlights vs. Crime**

Streetlight_Count has a moderate positive correlation with Crime_Count (0.56) and Arrest_Count (0.68), indicating that areas with more streetlights tend to have higher crime and arrest numbers. There is a weak positive correlation between Streetlight_Count and Night_Crime_Prop (0.20), suggesting a slight tendency for more streetlights to be associated with a higher proportion of nighttime crimes.

The correlation between Streetlight_Count and income variables (0.11 with MEDHINC_CY, 0.22 with AVGHINC_CY) is weak, implying that streetlight presence is not strongly tied to income levels.

- **Crime and Arrests**

Crime_Count and Arrest_Count are strongly positively correlated (0.97), indicating that areas with more reported crimes also tend to have more arrests. Night_Crime_Prop has a weak positive correlation with Crime_Count (0.044) and Arrest_Count (0.12), suggesting that the proportion of nighttime crimes does not strongly drive overall crime or arrest rates.

2.4 Exploratory Data Analysis Conclusion

Four things come out of the exploratory analysis. Larceny is the most common crime in every ward and every time period, and it is heaviest in Ward 3 and during the afternoon and evening, which is when the most people are out. Aggravated Assault and Robbery concentrate in Wards 3 and 5. Crime peaks from midday to early evening, and violent crime rises at night by a smaller amount, which is the gap streetlighting would have to close. Income correlates negatively with both crime and arrest counts, while streetlight count correlates positively with crime, so lighting on its own does not appear to suppress crime where the baseline is already high. The modeling in the next chapter tests the two hypotheses against these patterns.

Chapter 3

Results

3D Regression Surface: Crime Count vs Income & Streetlights

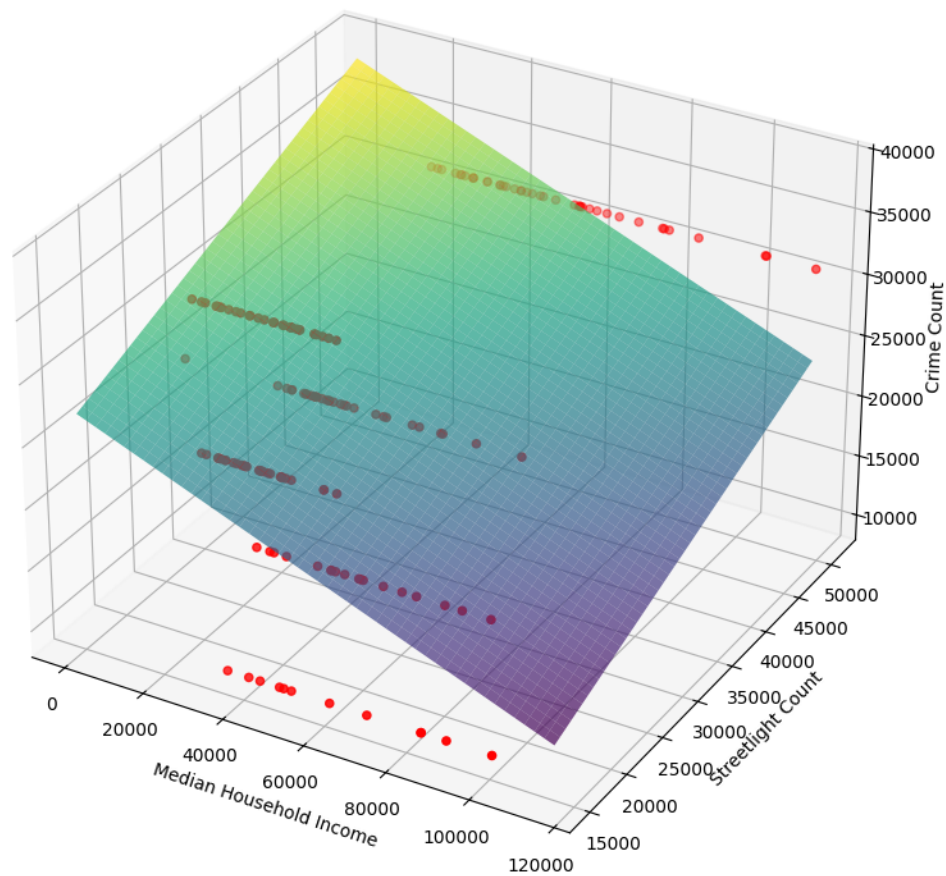


Figure 3.1: 3D Regression Surface

3.1 Ridge Regression

We applied Ridge Regression, seen in Figure 3.2, to analyze hourly crime counts in the East, Midtown, South, and West police divisions, using the hour of the day as the independent variable. The regularization penalty reduces model variance and guards against overfitting. The regression model was trained separately for each division, and we evaluated model performance using three metrics: R^2 score, Mean Absolute Error (MAE), and Mean Squared Error (MSE).

For each division, the model produced a linear equation of the form:

$$\text{CrimeCount} = \beta_0 + \beta_1(\text{Hour})$$

where the coefficients β_0 and β_1 varied by division. The slope coefficient was highest in the South Division, which has the most pronounced upward trend, meaning crime there climbs fastest as the day goes on. In contrast, Midtown and East had lower slope coefficients, suggesting a more moderate hourly increase.

The R^2 scores varied across divisions:

- **South Division** showed the strongest model fit with an R^2 value closest to 1, indicating that a large portion of the variance in crime count was explained by the hour of the day.
- **East and Midtown Divisions** had moderate R^2 scores, suggesting the model captured the general upward trend but not all variability.
- **West Division** exhibited the lowest R^2 score among the four, largely due to the higher variance in observed crime counts that the linear model could not fully account for.

MAE and MSE followed similar patterns. South Division had the lowest errors, reinforcing the strength of the model in that area. West Division had the highest error values, confirming the model's limited ability to account for the spread in crime counts, especially during peak hours.

Overall, Ridge Regression, seen in Figure 3.2, successfully modeled the general trend of increasing crime throughout the day. However, the variability in model performance across divisions suggests that linear models may not be sufficient for capturing more complex patterns, particularly in areas with higher variance. Future analyses could explore nonlinear models or include additional predictors (e.g., day of the week or crime type) to enhance predictive accuracy.

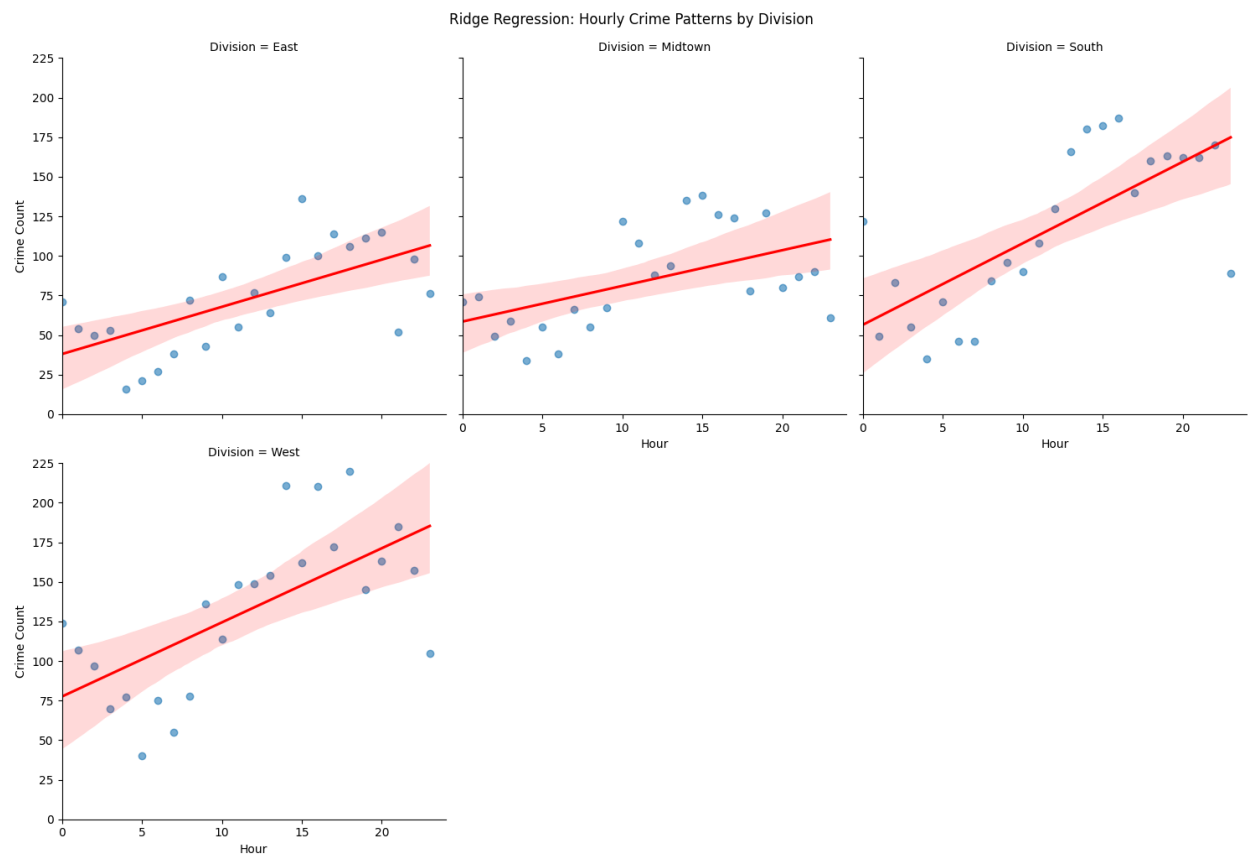


Figure 3.2: Ridge Regression

3.2 Model Performance Evaluation

The Random Forest (RF) and Logistic Regression (LR) models were evaluated to predict high-crime wards based on median household income (MEDHINC_CY), average household income (AVGHINC_CY), and streetlight presence (Streetlight_Count). Performance metrics including accuracy, F1-score, and detailed classification reports are presented below.

3.2.1 Random Forest Performance

This section implements RF and LR models to predict whether a ward has a high crime rate, using predictors related like income, streetlight presence, and crime timing. It also evaluates model performance and interprets which features are most important.

The Random Forest model achieved an accuracy of 0.97 and an F1-score of 0.95, indicating strong predictive performance. The classification report is summarized in Table 3.1

Table 3.1: Random Forest Classification Report

	Precision	Recall	F1-Score	Support
Class 0 (Low Crime)	1.00	0.95	0.98	22
Class 1 (High Crime)	0.91	1.00	0.95	10
Accuracy		0.97		32
Macro Avg	0.95	0.98	0.96	32
Weighted Avg	0.97	0.97	0.97	32

Feature importance analysis, seen in Figure 3.3 reveals Streetlight_Count as the most influential predictor (importance score 0.5), followed by MEDHINC_CY and AVGHINC_CY (0.25 each), suggesting that urban infrastructure plays a significant role in crime prediction, potentially more than socioeconomic factors. This partially supports the hypothesis that streetlight presence influences crime rates, though the positive correlation noted elsewhere suggests a reactive rather than preventive effect.

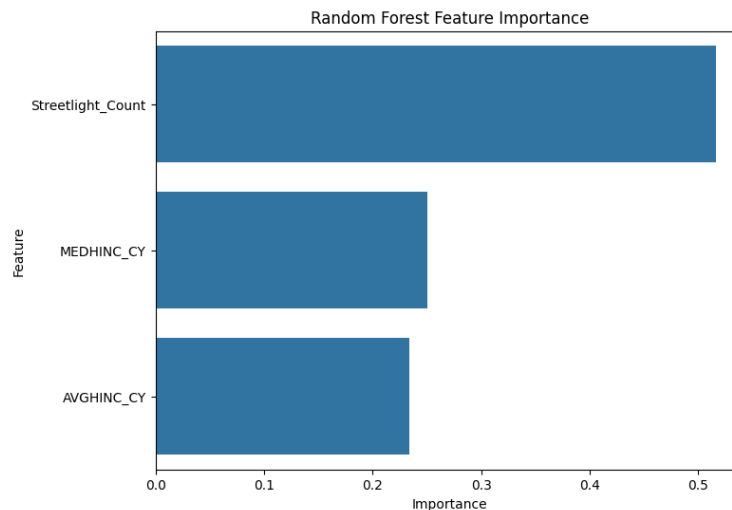


Figure 3.3: Feature Importance

3.2.2 Logistic Regression Performance

The Logistic Regression model yielded an accuracy of 0.75 and an F1-score of 0.71, performing less effectively than the Random Forest model. The classification report is detailed in Table 3.2.

Table 3.2: Logistic Regression Classification Report

	Precision	Recall	F1-Score	Support
Class 0 (Low Crime)	1.00	0.64	0.78	22
Class 1 (High Crime)	0.56	1.00	0.71	10
Accuracy		0.75		32
Macro Avg	0.78	0.82	0.75	32
Weighted Avg	0.86	0.75	0.76	32

3.2.3 Feature Importance Analysis

The Random Forest model’s feature importance analysis revealed that the number of streetlights (Streetlight_Count) was the most influential predictor of high-crime wards, with an importance score of approximately 0.5. Median household income (MEDHINC_CY) and average household income (AVGHINC_CY) followed, with importance scores around 0.25 each. This suggests that the presence of streetlights has a stronger association with crime rates compared to income levels, providing partial support for the hypothesis that streetlight presence influences crime rates. However, the income-related features still contribute meaningfully, indicating that wealth levels in neighborhoods also play a role in predicting crime, aligning with the hypothesis that thefts and violent crimes may vary between richer and poorer areas.

The Random Forest model outperformed Logistic Regression in predicting high-crime wards, with higher accuracy and F1-scores. The feature importance analysis highlights the significant role of streetlight presence in predicting crime rates, suggesting that urban infrastructure may influence crime more than socioeconomic factors like income in this dataset. These findings support the hypothesis that streetlight presence impacts crime rates and indicate a moderate relationship between crime and neighborhood wealth levels.

3.3 Regression Analysis Results

This section presents the results of two Ordinary Least Squares (OLS) regression models to analyze the impact of Median Household Income (MEDHINC_CY) and Streetlight Count (Streetlight_Count) on Crime Count. The models address the hypotheses: (1) whether thefts and violent crimes occur more often in richer or poorer neighborhoods, and (2) whether the presence of streetlights influences crime rates.

3.3.1 Model 1: Crime Count vs Median Household Income

The first OLS regression model, seen in Figure 3.4, examines the relationship between Crime Count and Median Household Income (MEDHINC_CY). The model summary is presented in Table 3.3.

Table 3.3: OLS Regression Summary

	Coefficient	Std. Error	t-value	P-value	95% Conf. Interval
Constant	30690	1244.395	24.662	0.000	[28240, 33140]
MEDHINC_CY	-0.1067	0.025	-4.199	0.000	[-0.157, -0.057]
R-squared: 0.101, Adjusted R-squared: 0.095					
F-statistic: 17.63, Prob (F-statistic): 4.48e-05					

The negative coefficient for MEDHINC_CY (-0.1067 , $p < 0.001$) indicates a statistically significant inverse relationship between median household income and crime count. As median income increases by one unit, crime count decreases by approximately 0.1067 units, supporting Hypothesis 1 that crime rates are higher in poorer neighborhoods. The R-squared value of 0.101 suggests that 10.1% of the variance in crime count is explained by median household income alone, indicating a modest but significant relationship. The scatterplot of Crime Count vs Median Household Income visually confirms this downward trend, showing higher crime counts at lower income levels.

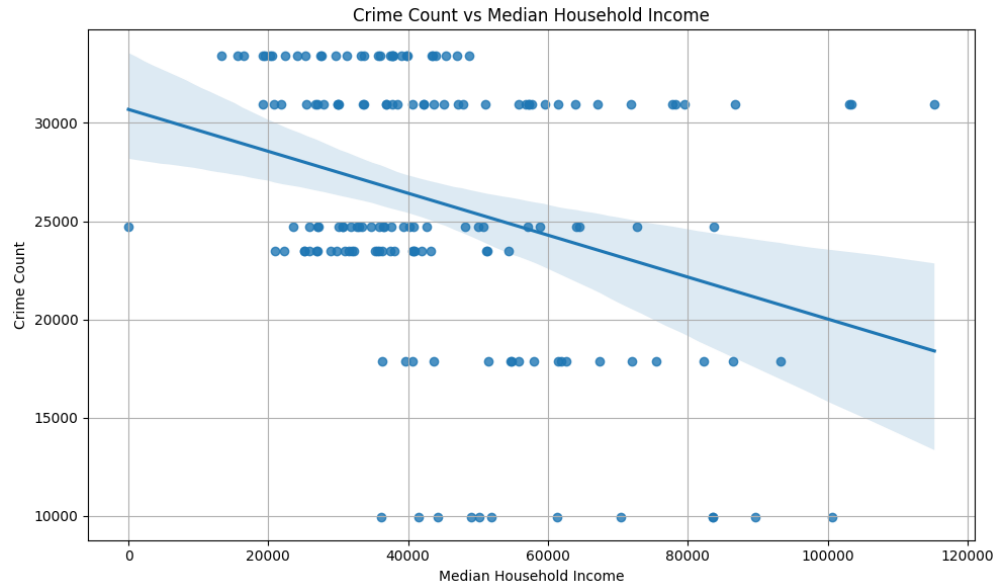


Figure 3.4: Crime Count vs Median Household Income

3.3.2 Model 2: Crime Count vs Median Household Income and Streetlight Count

The second OLS regression model, seen in Figure 3.1, includes both Median Household Income (MEDHINC_CY) and Streetlight Count as predictors of Crime Count. The model summary is shown in Table 3.4.

Table 3.4: OLS Regression Summary: Crime Count vs Median Household Income and Streetlight Count

	Coefficient	Std. Error	t-value	P-value	95% Conf. Interval
Constant	22560	1252.670	18.010	0.000	[20100, 25020]
Streetlight_Count	0.3069	0.030	10.205	0.000	[0.247, 0.366]
MEDHINC_CY	-0.1281	0.020	-6.454	0.000	[-0.167, -0.089]
R-squared: 0.461, Adjusted R-squared: 0.454					
F-statistic: 66.67, Prob (F-statistic): 1.18e-21					

This model shows improved explanatory power with an R^2 of 0.461, indicating that 46.1% of the variance in crime count is explained by the combination of median household income and streetlight count. Both predictors are statistically significant ($p < 0.001$). The coefficient for Streetlight_Count (0.3069) suggests a positive relationship with crime count, meaning that an increase in streetlights is associated with an increase in crime, which is contrary to Hypothesis 2, that streetlight presence reduces crime rates. The coefficient for MEDHINC_CY (-0.1281) remains negative and significant, reinforcing the finding from Model 1 that higher income is associated with lower crime counts. The 3D regression surface plot illustrates this relationship, showing a downward slope with respect to median household income and an upward slope with respect to streetlight count.

3.3.3 Discussion & Ethical Considerations

The regression analysis provides insights into the relationships between crime count, median household income, and streetlight presence. Model 1 confirms Hypothesis 1, showing that crime counts are higher in poorer neighborhoods, as evidenced by the significant negative coefficient for MEDHINC_CY. Model 2, however, challenges Hypothesis 2: while the Random Forest feature importance analysis (from the previous section) highlighted Streetlight_Count as the most influential predictor, the positive coefficient in the OLS model suggests that areas with more streetlights experience higher crime counts. This could indicate that streetlights are more commonly installed in high-crime areas as a response to crime, rather than as a deterrent. Income and streetlight count together explain 46.1% of the variance in Model 2 ($R^2 = 0.461$), against 10.1% for income alone, so neither factor accounts for these patterns by itself.

The analysis raises ethical concerns about data use, method, and application, and predictive policing is the sharpest of them. For example, while the datasets used offered valuable insights, they also presented biases when examined closely. Notably, there could be over-policing in lower-income areas, potentially skewing results and reinforcing stereotypes about poverty and crime. Likewise, the streetlight dataset assumes uniform effectiveness of lighting without accounting for factors like maintenance or perceived safety.

Furthermore, the correlation analysis and spatial joins can highlight associations, but they could also be misinterpreted as evidence of causation, such as a negative correlation between streetlight density and nighttime crime, leading to an overemphasis on lighting being the sole issue, rather than pre-existing social issues. Unfortunately, the use of machine learning models only exploits these connections, adding further reinforcement to existing disparities. Therefore, it is important to take heed that while our analysis highlights key connections between crime in Tucson with surrounding factors, there still exists a standard to uphold in the name of ethical rigor in crime data research. Transparency is key, and can help avoid reinforcing inequality among people, and cities as a whole.

Chapter 4

Conclusion

This study tested two hypotheses about crime in Tucson, Arizona: (1) that thefts and violent crimes occur more frequently in poorer neighborhoods than richer ones, and (2) that the presence of streetlights reduces crime rates. The first held. The second did not, and the data points the opposite way.

Our findings confirm Hypothesis 1, demonstrating a statistically significant inverse relationship between median household income and crime rates. Neighborhoods with lower median incomes, particularly in Wards 3 and 5, experience higher rates of both theft (e.g., Larceny) and violent crimes (e.g., Aggravated Assault and Robbery), as evidenced by the OLS regression results (Model 1, $R^2 = 0.101$) and the negative correlation between income and crime count (-0.32 for MEDHINC_CY). Hipp [Hip07] reports the same relationship between income inequality and neighborhood crime rates.

However, Hypothesis 2, which posited that streetlight presence reduces crime rates, was not supported. Both the OLS regression (Model 2, $R^2 = 0.461$) and Random Forest feature importance analysis (Streetlight_Count importance score of 0.55) revealed a positive association between streetlight count and crime rates, suggesting that streetlights are more prevalent in high-crime areas, possibly as a reactive measure rather than a preventive one. Welsh and Farrington [WF08] found that improved street lighting reduces certain types of crime, which this data does not reproduce. The weak positive correlation between streetlight count and nighttime crime proportion (0.20) further suggests that lighting alone may not significantly deter nighttime criminal activity in Tucson.

The exploratory data analysis provided additional context, revealing that Larceny dominates as the most prevalent crime across all wards and time periods, with peak crime hours occurring between 10 AM and 8 PM. This temporal pattern suggests that crime prevention strategies should prioritize daytime and early evening hours, potentially through increased community policing or public awareness campaigns. The strong correlation between crime and arrest counts (0.97) indicates that policing efforts are closely aligned with reported crime, but raises ethical concerns about potential over-policing in lower-income neighborhoods, which could perpetuate systemic biases.

The Random Forest model predicted high-crime wards more accurately than Logistic Regression, 0.97 against 0.75. That accuracy is a reason for caution rather than confidence: a model this good at reproducing the arrest record is also good at reproducing whatever bias produced it. Two limitations bear directly on that. Crime reporting rates are not uniform across wards, and the streetlight records carry no maintenance history, so a listed light is not necessarily a working one.

4.0.1 Future Work

Future research could build on these findings by incorporating additional variables, such as education levels, unemployment rates, or housing quality, to further elucidate the socioeconomic drivers of crime. Expanding the temporal scope to include seasonal or day-of-week variations could reveal more nuanced patterns in crime distribution. Additionally, qualitative studies, such as community surveys or interviews, could provide insights into residents' perceptions of safety and the effectiveness of streetlights, addressing limitations in the current dataset. Randomized streetlight installation across comparable areas would test the causal claim directly, which no amount of further work on this observational data can do. Finally, integrating demographic

data on race and ethnicity could help assess whether arrest disparities reflect systemic biases, informing more equitable policing strategies.

The clearest result is the one that contradicted us. We expected streetlights to suppress crime and found them concentrated where crime already was, which is what installing lights in response to crime would look like. Observational data cannot separate those two explanations, and this report does not claim to.

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